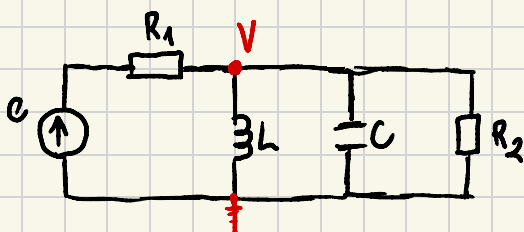


ĆWICZENIA 8

k.3 z.2

Zrób bilans mocy



$$e(t) = 10\sqrt{2} \sin(\omega t + 90^\circ)$$

$$R_1 = 2 \Omega \quad R_2 = 1 \Omega$$

$$\omega L = 1 \Omega \quad \frac{1}{\omega C} = 2 \Omega$$

$$E = j10 \text{ [V]}$$

Wierzymy prądy

$$I_L = \frac{V}{Z_L} = \frac{-1+j3}{j} = \frac{-j-3}{-1} = 3+j$$

$$I_C = \frac{V}{Z_C} = \frac{-1+j3}{-j2} = \frac{1-j3}{j2} \cdot \frac{j2}{j2} = \frac{j2-6}{-4} =$$

$$= -1,5 - j0,5$$

$$I_{R2} = \frac{-1+j3}{1} = -1+j3$$

KOŁOS 1 (3 zadania)

- metoda oczekowa

- metoda potencjałów

- THEVENIN

$$V\left(\frac{1}{R_1} + \frac{1}{Z_L} + \frac{1}{Z_C} + \frac{1}{R_2}\right) = \frac{E}{R_1}$$

$$V\left(\frac{1}{2} + \frac{1}{j} + \frac{1}{-j2} + \frac{1}{1}\right) = \frac{j10}{2}$$

$$V\left(\frac{3}{2} - j + j\frac{1}{2}\right) = j5$$

$$V\left(\frac{3}{2} - j\frac{1}{2}\right) = j5$$

$$V = \frac{j5}{1,5 - j0,5} \cdot \frac{1,5 + j0,5}{1,5 + j0,5}$$

$$V = \frac{j7,5 - 2,5}{2,25 + 0,25} = -1 + j3$$

Liery I_{R1}

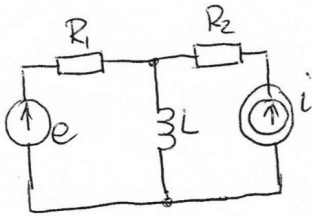
$$E - R_1 \cdot I_{R1} = V$$

$$R_1 \cdot I_{R1} = E - V$$

$$I_{R1} = \frac{E - V}{R_1} = \frac{j10 - (-1 + j3)}{2} = \frac{j10 + 1 - j3}{2} = 0,5 + j3,5$$

k.3. 2.2 c)

c)



$$R_1 = 10\Omega, \quad R_2 = 20\Omega, \quad L = 2H$$

$$e(t) = 20\sqrt{2} \sin(5t + 90^\circ)$$

$$i(t) = 2 \sin(5t + 45^\circ)$$

$$X_L = \omega \cdot L = 5 \cdot 2 = 10\Omega$$

$$Z_L = j10\Omega$$

$$E = j20$$

$$I = \frac{2}{\sqrt{2}}(\cos 45^\circ + j \sin 45^\circ) = \frac{2}{\sqrt{2}}\left(\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}\right) = 1 + j$$

$$V\left(\frac{1}{R_1} + \frac{1}{Z_L} + \frac{1}{R_2}\right) = \frac{E}{R_1} + I$$

$$V\left(0,1 + \frac{1}{j10}\right) = \frac{j20}{10} + 1 + j$$

$$V(0,1 - j0,1) = 1 + j3$$

$$V = \frac{1 + j3}{0,1 - j0,1} = \frac{(1 + j3)(0,1 + j0,1)}{0,01 + j0,01} = \frac{0,1 + j0,1 + j0,3 - 0,3}{0,02} = \frac{-0,2 + j0,4}{0,02} = -10 + j20$$

$$I_L = \frac{V}{Z_L} = \frac{-10 + j20}{j10} = \frac{-10j - 20}{-10} = 2 + j$$

$$E - R_1 \cdot I_{R1} - V = 0$$

$$I_{R1} = \frac{E - V}{R_1} = \frac{j20 - (-10 + j20)}{10} = \frac{10}{10} = 1$$

$$I_{R2} = I = 1 + j$$

$$I_{R1} = 1 \quad |I_{R1}|^2 = 1^2 + 0^2 \quad P_{R1} = \overset{||I_{R1}||^2}{1} \cdot 10 = 10W$$

$$I_{R2} = 1 + j \quad |I_{R2}|^2 = 1^2 + 1^2 \quad P_{R2} = 2 \cdot 20 = 40W$$

$$I_L = 2 + j \quad |I_L|^2 = 2^2 + 1^2 \quad Q_L = |I_L|^2 \cdot X_L = 5 \cdot 10 = 50W$$

$$S_E = E \cdot I_{R1}^* = j20 \cdot (1) = j20$$

$$S_I = V_{Zr} \cdot I^* = (10 + j40) \cdot (1 - j) = 10 - j10 + j40 + 40 = 50 + j30$$

$$V + R_2 \cdot I - V_{Zr} = 0$$

$$V_{Zr} = V + R_2 \cdot I$$

$$V_{Zr} = -10 + j20 + 20(1 + j) = 10 + j40 [V]$$

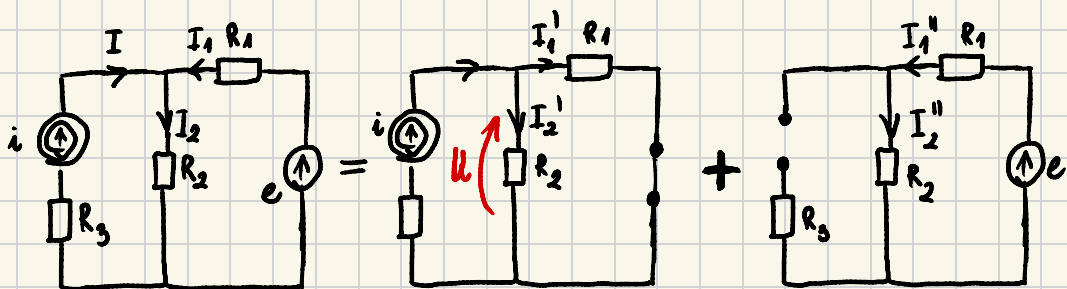
$$S_g = S_E + S_I = j20 + 50 + j30$$

$$S_g = 50 + j50$$

$$P_{\text{odtr}} = P_{R1} + P_{R2} = 10 + 40 = 50W$$

$$Q_L = 50$$

Zad. 3



$$i(t) = 2A$$

$$e(t) = 10V$$

$$R_1 = 6\Omega$$

$$R_2 = 4\Omega$$

$$R_3 = 10\Omega$$

$$R_{12} = \frac{6 \cdot 4}{6 + 4} = 2,4\Omega$$

$$U = R_{12} \cdot I = 2,4 \cdot 2 = 4,8 [V]$$

$$I_2' = \frac{U}{R_2} = \frac{4,8}{4} = 1,2 [A]$$

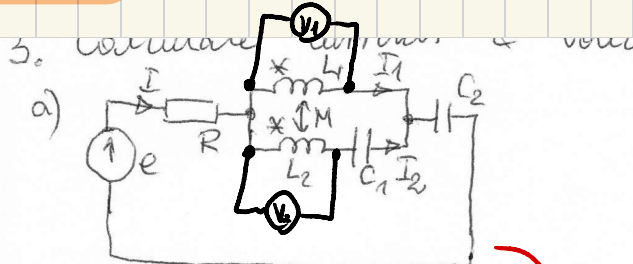
$$I_1' = \frac{U}{R_1} = \frac{4,8}{6} = 0,8 [A]$$

$$I_1'' = I_2'' = \frac{E}{R_1 + R_2} = \frac{10}{10} = 1 [A]$$

$$I_1 = -I_1' + I_1'' = -0,8 + 1 = 0,2 [A]$$

$$I_2 = I_2' + I_2'' = 1 + 1,2 = 2,2 [A]$$

Zad. 4 K.6 Z.3



$$e(t) = 20\sqrt{2} \sin(\omega t + 90^\circ)$$

$$\omega L_1 = 20\Omega, \omega L_2 = 30\Omega, \omega M = 10\Omega$$

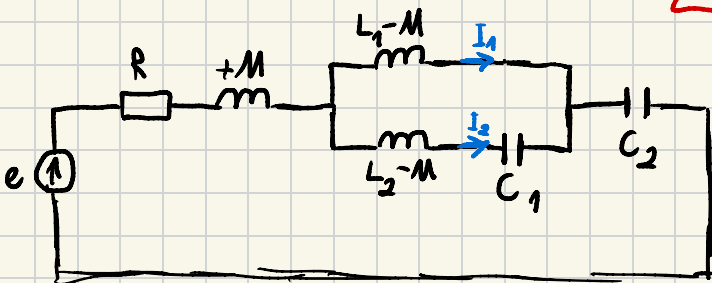
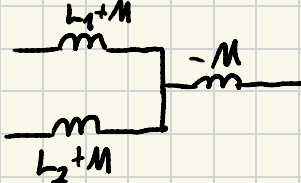
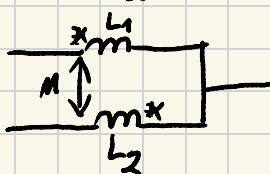
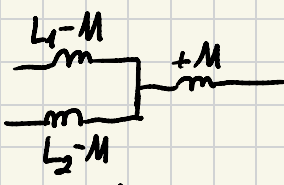
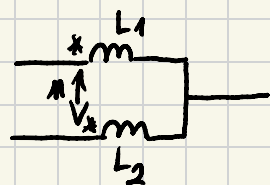
$$R = 20, \frac{1}{\omega C_1} = 20\Omega, \frac{1}{\omega C_2} = 30\Omega$$

$$U_{L1} = jX_{L1} \cdot I_1 + jX_M \cdot I_2$$

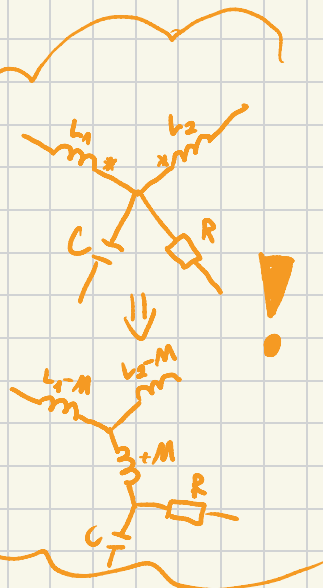
Jesli $\vec{I}_1, \vec{I}_2$ sa ten sam kierunku to "+"

W przeciwnym przeciwnych to "-"

$$E = j20 \quad Z_{L1} = j20 \quad Z_{L2} = j30$$



Zderzy od kierunku prądów



$$Z_{L1M} = Z_{L1} - Z_M = j20 - j10 = j10 \Omega$$

$$Z_{L2M} = Z_{L2} - Z_M + Z_C = j30 - j10 - j20 = 0 \Omega \quad \leftarrow \text{Rezonans napięcia}$$

$$\Downarrow$$

$$I_1 = 0$$

$$Z_2 = R + j\omega M - jX_{C2} = 20 + j10 - j30 = 20 - j20$$

$$I = \frac{E}{Z_2} = \frac{j20}{20 - j20} = 0,5 + j0,5$$

$$I_2 = I$$

$$U_{L2} = jX_{L2} \cdot I_2 + jX_M \cdot I_1 = jX_{L2} \cdot I_2 = u + jv \rightarrow (V_2) = \sqrt{u^2 + v^2}$$

$$U_{L1} = jX_{L1} I_1 + jX_M \cdot I_2 = 0 + jX_M \cdot I_2 = c + jd \rightarrow$$